

Ibraheem Mohammed AlHabib

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Education

King Abdullah University of Science and Technology (KAUST)

Thuwal, Saudi Arabia

Master of Science in Computer Science

Aug 2023 - May 2025

- Relevant Coursework: Deep Learning for Visual Computing, Deep Generative Modeling, Computing Systems and Concurrency, Advanced Distributed & Networked Systems

University of North Carolina, Chapel Hill

Chapel Hill, NC

Bachelor of Science in Computer Science

Sept 2019 - May 2023

Experience

King Abdullah University of Science and Technology

Thuwal, Saudi Arabia

Research Assistant, SANDS Lab

May 2024 - Dec 2024

- Developed ARCA (Agentic Root Cause Analyzer), an intelligent system that automates root cause analysis of service-level objective violations in microservice architectures and identifies root causes in 60% of memory degradation and network degradation scenarios.
- Engineered CodeReAct architecture combining adaptive investigation patterns with dynamic code execution, achieving 93% accuracy in system understanding.
- Built comprehensive observability platform integrating Prometheus, Docker, and Jaeger with automated fault injection capabilities for benchmarking and evaluation of diagnostic performance.

KGSP Alumni Council

Saudi Arabia

Outreach Chair

Nov 2023 - Nov 2024

- Spearheaded expansion of alumni network to major Saudi cities including Riyadh and Jeddah by coordinating and hosting strategic networking events.
- Developed and implemented multi-channel outreach strategies that significantly increased attendance in alumni events by 46 percent from the prior term.

Personal Projects

AI-powered Emotion-Based Image Editing

- Engineered FastAPI backend service with multi-stage AI pipeline integrating Claude's vision capabilities and DALL-E's image generation to transform images based on specified emotions.
- Implemented production-ready API with input validation, caching, rate limiting, and async processing.
- Developed advanced prompt engineering strategies and robust error-handling system.

Deep Learning for Personalized Movie Recommendations

- Developed recommendation system with hybrid neural architecture using Deep Factorization Machine (DeepFM) and performed extensive hyperparameter tuning to optimize performance.
- Engineered and processed dataset of 32M+ entries from MovieLens and IMDB, optimizing with mixed precision training and memory-efficient data loading in PyTorch.
- Implemented metadata-driven embeddings and user profiling techniques that addressed cold-start problem, with 83% of predictions falling within 1.0 stars of actual ratings.

Full-Stack Web Application with Cloud Deployment

- Developed React/FastAPI portfolio with CI/CD pipeline, containerized Docker backend, and secure Azure deployment featuring custom domain integration and HTTPS/CORS security configurations.
- Integrated the Affective Image Modification project as an interactive demo, and currently working on incorporating the Movie Recommendation System for a comprehensive showcase of technical capabilities.

Skills

Programming	Python, Java, C, Go, SQL
Python Libraries	PyTorch, Pandas, NumPy, SciPy, Hugging Face Transformers
Web & Backend	React, FastAPI, Docker, Azure
DevOps & Tools	Git, CI/CD (Continuous Integration/Deployment)
Deep Learning	Neural Architectures, Computer Vision, Recommendation Systems
LLM Engineering	Prompt Engineering, Agent Design, DSPy, Claude Integration